



Proposal by **Glitchmore Girls**





MEET THE TEAM



Disha Kukkal

Hi, I am Disha, a B.Tech student at IIIT Delhi (CSSS), fascinated by the intersection of quantum computing, AI, and data science. Part of a Quantum Computing program by The Coding School (IBM Quantum), mentored by MIT and UC Berkeley faculty I thrive on transforming abstract ideas into practical, impactful solutions.



Yashswinie Arya

Hi, I am Yashswini, a B.Tech EVE student focused on electronics, embedded systems, and hardware design. I have experience working with microcontrollers, PCB design, and real-time systems, and have built a wall-climbing cleaning robot while also exploring RF-based device research. I enjoy designing practical, hardware-driven solutions.



Varshamegana Atmakuri

Hi, I am Varsha, a B.Tech student at IIIT Delhi majoring in CSSS. I have worked as a Content Lead at Meraki ArtSoc and a core member of CyFuse, while also contributing as an undergraduate researcher in Human-Centered Design, where I worked on data collection and analysis. I enjoy combining creativity with structured problem-solving to build user-focused solutions.





PROBLEM STATEMENT



Problem Statement

Neurodivergent children struggle with correct letter formation and stroke direction due to lack of real-time guidance.

Problem Statement Expansion

Current learning methods rely on delayed correction from teachers or parents, which is often inconsistent and insufficient.

Children repeatedly practice incorrect writing patterns, reinforcing errors like letter reversals (b/d, p/q) and improper stroke order, making correction harder over time.

How it manifests for the user or client

Children develop incorrect writing habits early
Frequent mistakes in stroke direction and letter formation
Dependence on constant supervision for correction
Frustration and reduced confidence during learning

How it impacts the user or client

For Children:

Slower learning and poor handwriting development
Increased frustration and reduced self-confidence
Difficulty keeping up academically

For Parents & Educators:

Continuous need for manual correction
Time-consuming supervision
Lack of scalable, personalized learning support



Project Details

Smart Pencil Gripper - NeuroFlow

A compact smart pencil gripper that helps children, especially neurodivergent learners, improve writing using real-time feedback. It uses IMU and pressure sensors to detect stroke direction and provides immediate haptic correction. The system integrates with an application to track progress and enable adaptive learning, helping children improve writing accuracy, confidence, and independence.

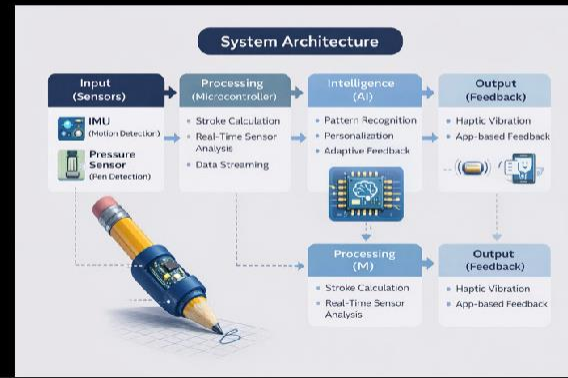
Demographic it Impacts:

Neurodivergent children (ages 4–9) — dyslexia, dysgraphia

Early-stage learners developing writing skills

Parents supporting home learning

Teachers and special educators





User Persona Mindset



Build for neurodivergent children with real learning challenges

Focus on motor skill development, not just academic output

Design for real-time guidance, not delayed correction

Assume limited supervision at home and in classrooms

Prioritize simplicity, comfort, and child-friendly interaction

Create solutions that build confidence, not frustration



AARAV

Primary school student
Diagnosed with dyslexia /
dysgraphia
Early-stage writer
6 years old

USER PERSONA 1



ABOUT THE USER

Aarav is a young student learning how to write alphabets and numbers. He is curious and eager to learn but often struggles with writing correctly, especially confusing letters like b/d and p/q. He is still developing motor control and needs guidance while practicing.

PROBLEMS

- Frequent letter reversals (b/d, p/q, 6/9)
- Incorrect stroke direction and formation
- Delayed feedback from teachers/parents
- Loss of confidence due to repeated mistakes

CHALLENGES

- Maintaining proper writing flow
- Understanding correct stroke order
- Building confidence while learning
- Practicing independently

ABOUT THE USER

- Curious
- Easily distracted
- Learner
- Needs guidance
- Sensitive to feedback

GOALS AND NEEDS

- Learn correct writing habits
- Receive real-time guidance
- Build confidence
- Enjoy learning without pressure



NEHA

- 34 years old
- Working parent
- Mother of a neurodivergent child (dyslexia)
- Balances work + child's learning needs

USER PERSONA 2



ABOUT THE USER

Neha is a working parent of a neurodivergent child who struggles with writing due to dyslexia and dysgraphia. She actively tries to support her child's learning but finds it difficult to provide constant supervision or correct writing techniques effectively. She wants her child to learn without stress but often feels overwhelmed balancing guidance and encouragement

PROBLEMS

- Cannot monitor writing practice constantly
- Unsure how to correct stroke formation
- Child gets frustrated during practice
- Limited access to specialized support

CHALLENGES

- Balancing work and child's learning
- Providing consistent guidance
- Avoiding pressure on the child
- Ensuring effective learning at home

ABOUT THE USER

- Supportive and involved
- Time-constrained
- Emotionally invested
- Not trained in teaching methods
- Seeks effective learning tools

GOALS AND NEEDS

- Simple, guided learning tools
- Reduced need for constant supervision
- Stress-free learning environment
- Visible improvement in child's writing



USER PERSONA 3



MR SHARMA

- 42 Years old
- Primary School Teacher/Special kids educator
- Works with neurodivergent students

ABOUT THE USER

Mr. Sharma is a primary school teacher who works with multiple students, including neurodivergent learners with dyslexia and dysgraphia. He understands the importance of correct stroke formation but often finds it difficult to provide real-time, individualized attention to every student. He needs tools that can support his teaching and ensure consistent guidance even when he cannot intervene directly.

PROBLEMS

- Cannot give one-on-one attention continuously
- Difficult to correct mistakes in real-time
- Limited classroom time
- Repetitive correction process

CHALLENGES

- Managing diverse learning needs
- Scaling personalized guidance
- Maintaining student engagement
- Tracking individual progress

ABOUT THE USER

- Experienced educator
- Patient and empathetic
- Handles multiple students
- Limited time per child
- Focused on effective learning

GOALS AND NEEDS

- Scalable learning support tools
- Real-time correction assistance
- Efficient teaching methods
- Better learning outcomes



Value proposition



Dyslexia Optimization: Uses the Lexend font and widened letter spacing to reduce visual crowding and improve reading accuracy.

Bite-Sized Pacing: Visual progress bars (e.g., "1 of 5") ensure tasks never feel overwhelming.

Actionable Dashboards: Educator views highlight class-wide error trends (e.g., stroke direction issues) to inform teaching, while Parent views offer specific, low-pressure home integration tips.

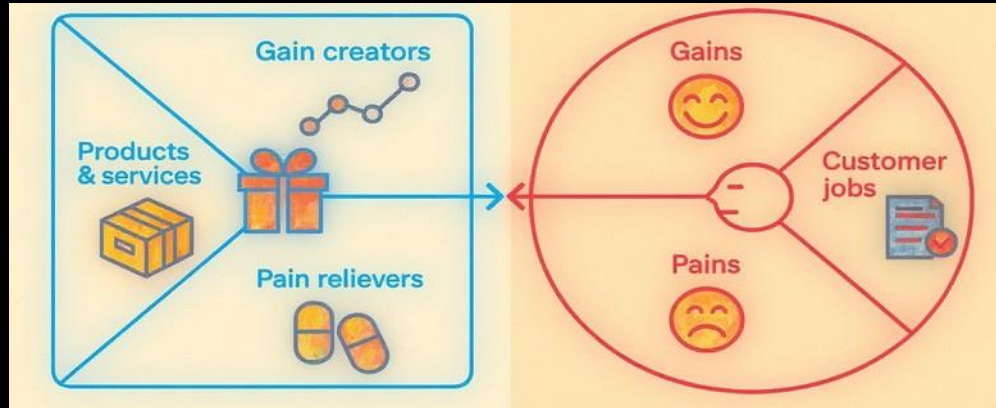
Building confidence in writing and motor skills.

A calm, focused, and adaptable learning environment tailored to their specific neurodivergent needs.

Caregivers gaining real-time, actionable insights into exactly where the child is struggling (e.g., specific letter mix-ups like b/d).

NeuroFlow Smart Pen: A hardware grip featuring an MPU6050 motion sensor and Arduino Nano.

NeuroFlow App Ecosystem: An adaptive digital interface featuring specialized learning modules and dedicated Parent/Educator analytical dashboards.



* Child: Learning to write legibly, developing fine motor control, and completing literacy/numeracy tasks.

Parents/Educators: Tracking the child's learning progress and providing effective support without causing stress.

Haptic Feedback: The smart pen uses a vibration motor to gently correct unstable writing motion in real-time, eliminating the need for stressful verbal correction.

Sensory-Friendly UI Modes: "ADHD Focus" mode turns off distracting animations/sounds and uses a calming blue palette. "Sensory Sensitive" mode dims contrast to prevent overstimulation.

No Mandatory Homework: Reduces anxiety by replacing traditional homework with low-pressure, playful activities for parents to do with their kids.

Severe anxiety, frustration, and pressure associated with traditional handwriting correction (e.g., verbal corrections or red marks).

Sensory overload (visual noise, unexpected sounds) from standard educational apps.

Teachers and traditional grading only evaluate the final written output, missing the actual motor struggles and unstable motions happening during the writing process.

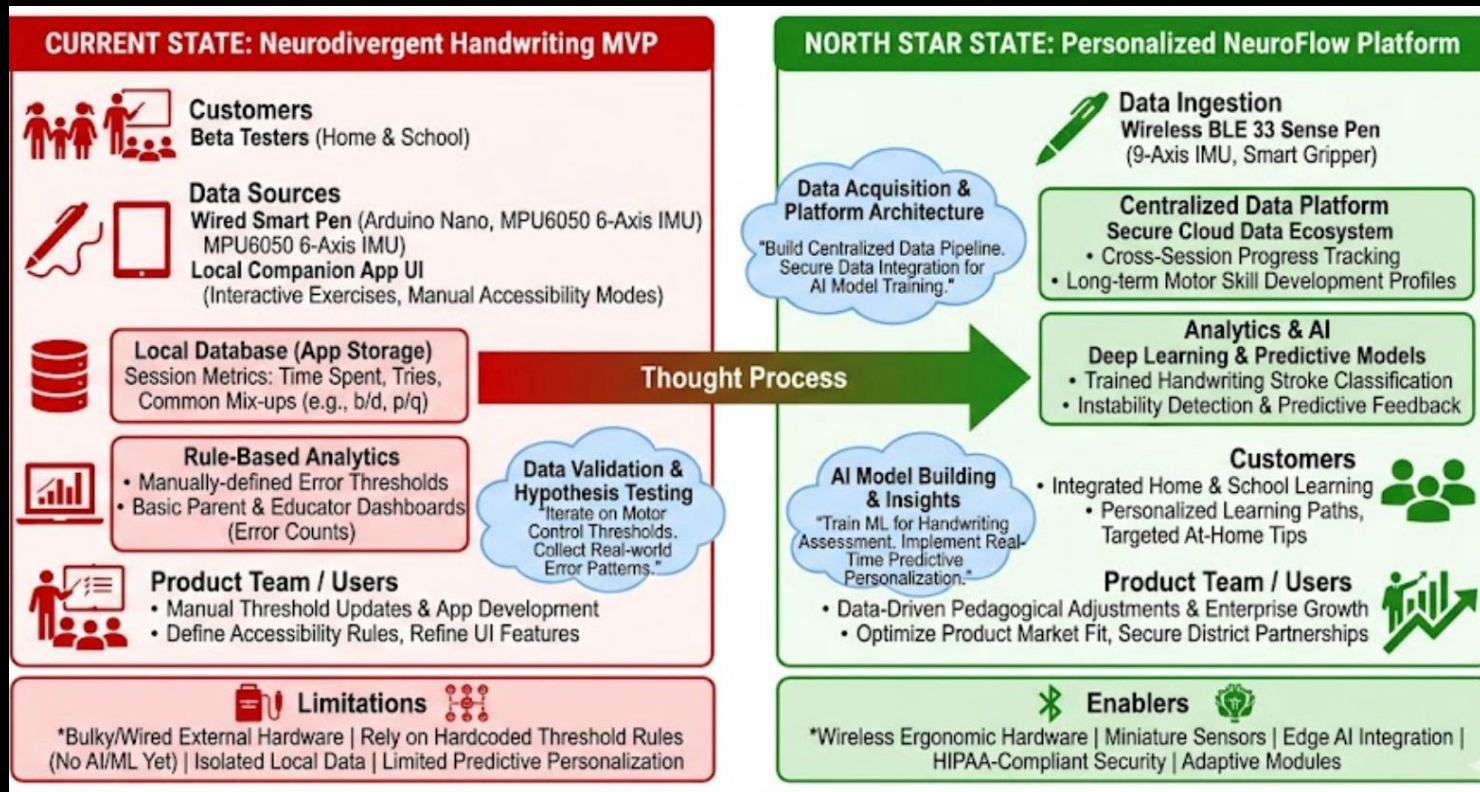


Thinking BIG

Current State vs Future State of our solution



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Architecture Discovery



Assumptions

- BLE 33 Sense accurately captures motor data.
- Kids respond well to haptics.
- Motion data reveals motor challenges.

Key Decisions



- Use integrated BLE 33 Sense pen.
- Add Dyslexia & ADHD modes; no homework.
- Train ML on motion datasets.

Team Alignment



Risk & Mitigation

- Risk: Bulky pen -> Mitigation: Ergonomic grip.
- Risk: Vibration startles -> Mitigation: Adjustable intensity.

Dependencies



- Stable Bluetooth & Cloud sync.
- Access to neurodivergent beta testers.
- Large motion datasets for ML.



Process & Data Flow



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FROM CUSTOMERS (Data In)

This section defines the inputs your system receives from the child and the hardware.

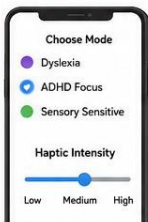
User Signals



MPU6050

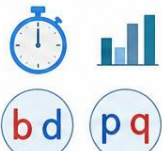
Continuous X, Y, and Z-axis acceleration and motion patterns captured in real-time by the MPU6050 sensor.

Interactions



User selections in the app, such as choosing specific accessibility modes (Dyslexia, ADHD Focus, Sensory Sensitive) and adjusting haptic intensity preferences.

Feedback & Events



Session metrics including time spent, number of tries, and specific stroke errors (e.g., mixing up b/d or p/q).

INTELLIGENT PROCESSING (The System Core)

How the data is handled internally.



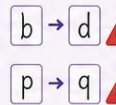
1. Input Processing

The Arduino Nano micro-controller processes the raw incoming sensor values and evaluates the immediate motion consistency.



2. Reasoning Engine (AI/ML)

A Machine Learning model, trained on extensive handwriting motion datasets, analyzes the child's movement patterns to detect instability, predict stroke direction errors, and determine the need for intervention.



3. Execution Engine

Based on the AI's assessment, the system triggers the physical vibration motor in the pen. It simultaneously updates the app's visual UI to guide the child's attention back to the correct stroke pattern.



4. Knowledge Base / Storage

The system securely logs this session data to build long-term personalized handwriting behavior profiles.



TO CUSTOMERS (Data Out)

Actionable outputs delivered back to the child, parents, and educators.

Automated Actions / Alerts



Immediate, gentle haptic feedback (vibration) directly from the pen to physically correct the child's stroke without verbal pressure.

Answers & Responses (Real-Time App Feedback)



Real-time visual feedback on the companion app, providing low-pressure corrections.

Dashboards & Reports

Parent View

- Daily practice time
- Attempt counts
- Specific letter mix-ups



Top Mix-Ups: b/d, p/q

Educator View

Class-wide tracking and trend analysis showing the most common error patterns across all students.



Recommendations



Draw letters in sand



Point out shapes in books



The system pushes "Optional ideas for parents," suggesting low-pressure, playful motor activities to try at home.



Rollout Roadmap



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PHASE 1	Feature Focus	Key Activities	Value Added
MVP – Validate & Learn 0-3 Months	Refine the current wired MPU6050 and Arduino Nano prototype.	Test the companion app's UI accessibility modes, such as ADHD Focus and Sensory Sensitive.	Validates that the vibration motor provides effective haptic feedback for correcting handwriting motion.
PHASE 2	Feature Focus	Key Activities	Value Added
Scale – Expand & Optimize 3-9 Months	Transition to integrated wireless hardware using the Arduino Nano BLE 33 Sense.	Embed the micro-controller directly inside a sleek, lightweight, and ergonomic pencil grip.	Removes bulky wiring, allowing children to write comfortably while the system collects real-time X, Y, and Z-axis acceleration and motion patterns.
PHASE 3	Feature Focus	Key Activities	Value Added
Optimize – Integrate & Automate 9-18 Months	Introduce robust Machine Learning models.	Train ML models on the collected handwriting motion datasets to detect instability and predict improvements.	Shifts from basic tracking to personalized handwriting behavior analysis, enabling detailed "Stability Over Time" tracking on the personalized AI dashboards.
PHASE 4	Feature Focus	Key Activities	Value Added
North Star – Innovate & Transform 18+ Months	Establish a complete ecosystem featuring a Smart Pen, Companion App, and Cloud Sync.	Function as an intelligent assistive learning tool that provides adaptive real-time handwriting analysis.	Empowers neurodivergent children with confidence while providing long-term insights and progress reports for continuous support.



Solution Inclusivity, Fairness and Explainability (1/2)



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1. Accessibility

- **Language:** UI designed for easy global localization.
- **Digital Access:** Runs on standard tablets, uses low-power BLE.
- **Disability:** Built for neurodiversity; features Dyslexia, ADHD Focus, and Low Vision modes.

YES

YES

YES



2. Representative Data Strategy

- **YES** — Data acquisition prevents bias.
- ML trained on continuous motion datasets, not static images.
- Partnering with inclusive programs for balanced neurodivergent baselines.



3. Target Demographics for Upcoming Pilot

	Neurodiversity	70% Neurodivergent, 30% Neurotypical
	Gender	50/50 Balanced split
	Income	Minimum 40% from Title I / underfunded public schools



Target: 50+ students across 3 diverse districts

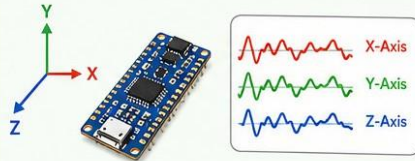


Solution Inclusivity, Fairness and Explainability (2/2)



Diverse Motion Data

ML models are trained using continuous motion datasets (X,Y,Z acceleration), not static images, to ensure inclusive representation for ADHD, Dyslexia, and Dysgraphia.



Diverse by ability, background, and learning style



Transparent & Traceable

No 'black box' AI. Outputs provide clear, explainable metrics (e.g., stroke errors) on dashboards. Immediate haptic feedback is a direct, understandable response.



Clear data. Explainable AI. Actionable insights.



Minimum Necessary Data

System only collects relevant handwriting patterns. No camera or microphone data is accessed. Focus is exclusively on fine motor skill validation.



No Camera



No Microphone



Handwriting Motion (X, Y, Z Acceleration)



Stroke Patterns & Pressure



Fine Motor Skill Validation



Privacy by design. Purpose by default.



Actual Solution Images/ Videos



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[Click Here for the Prototype video demo](#)

in case the click dosen't work:

<https://drive.google.com/file/d/1XyroXN2GIBvgFwo437p1jnajHVyJOb89/view?usp=sharing>



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THANK YOU

And all the best for
your solution judging

